Lab-03: Recognise Malware and Endpoint Warning Signs

Course: Cyber Security Awareness Workshop

Course code: C506

Time: 30 minutes

Alignment: LO1, LO3

Tool: [Cybersecurity Simulator](https://alfredang.github.io/cybersecuritysimulator/)

Scenario

Endpoint alerts reveal suspicious processes, persistence and outbound connections on an employee laptop.

Safety and authorisation boundary

- Use only the named training tool and the mock files in this folder.
- Do not enter real passwords, personal data, payment details or confidential information.
- Do not test any other website, network, account or organisation.
- Stop immediately if the target, data or behaviour differs from the written scope.
- Prefer reversible actions and preserve synthetic evidence.

Mock data

- `mock-endpoint-alerts.json`

Detailed procedure

1. Review each endpoint alert without opening or executing any referenced file.

2. Map the observable behaviour to a likely malware category: ransomware, trojan, spyware, worm, bot or rootkit.

3. Separate evidence from inference; identify what additional safe telemetry would confirm the hypothesis.

4. Build a short attack chain from initial access through execution, persistence and command-and-control.

5. Choose containment actions that preserve evidence and avoid spreading the infection.

6. Prepare a concise escalation note using who, what, when, where, impact and action already taken.

Completion standard

- Malware classification
- Attack-chain map
- Evidence vs inference notes
- Containment decision
- Escalation note

Reflection

What evidence would make you reverse your decision, and who would need to know?